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provisions for regulating the use of thermal waste treatment devices as equivalent means of compliance in accordance with regulation 4 of MARPOL Annex VI;
3 REQUESTS Parties to MARPOL Annex VI and other Member Governments to bring the 2023 TWTD Guidelines to the attention of shipowners, ship operators, shipbuilders, marine equipment manufacturers and any other interested groups;
4 AGREES to keep these Guidelines under review in light of experience gained with their application.
RESOLUTION MEPC.373(80) (adopted on 7 July 2023)
2023 GUIDELINES FOR THERMAL WASTE TREATMENT DEVICES (TWTD)MEPC 80/17/Add.1
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ANNEX

2023 GUIDELINES FOR THERMAL WASTE TREATMENT DEVICES (TWTD)

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1 Introduction

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Annex Form of TWTD Certificate

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1 Introduction

1.1 These Guidelines cover the approval, certification and in -service controls applicable to thermal waste treatment devices (TWTD) as equivalent means, under regulation 4 of MARPOL Annex VI, to incinerators as covered by regulation 16 of that Annex and as specifically provided for by paragraphs 1 and 5.2 of that regulation.

1.2 These Guidelines, as directed by PPR 6, are written on the basis of a technology-neutral, goal-based approach that can be applied to any thermal waste treatment device using, for example, gasification, hydrothermal carbonization, pyrolysis, plasma or other thermal means for the disposal of permitted garbage and other shipboard wastes generated during a ship's normal service.

1.3 As an alternative to conventional incinerators, as a means of disposal of garbage and other shipboard wastes, these thermal waste treatment devices remain subject to the same

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during a ship's normal service.

1.3 As an alternative to conventional incinerators, as a means of disposal of garbage and other shipboard wastes, these thermal waste treatment devices remain subject to the same prohibitions as to those materials which are not to be so disposed of as given in regulation 16.2 of MARPOL Annex VI.

1.4 A TWTD certified in accordance with these Guidelines should meet Performance Level 1 in terms of emissions to air which is comparable to the emission limit requirements given in the 2014 Standard specification for shipboard incinerators (resolution MEPC.244(66), as amended) this Performance Level should be demonstrated by in -service emission measurements. Where there is a related water discharge to sea, that also should be controlled as given in these Guidelines.

1.5 Additionally, an applicant may request certification to Performance Level 2. In that case not only should the Performance Level 1 requirements be met but there are detailed additional testing requirements which should be met prior to approval as an equivalent means together with tighter in-service emission limits.

2 General Basis of these Guidelines

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additional testing requirements which should be met prior to approval as an equivalent means together with tighter in-service emission limits.

2 General Basis of these Guidelines

2.1 In order to be "technology-neutral" these Guidelines follow a goal -based approach, the basis of which is:

.1 the in-service monitoring and recording of specified emissions;

.2 the identification of relevant Functional Objectives; and

.3 the applicant -proposed/demonstrated resolution of each Functional Objective by means of an applicant -compiled Thermal Waste Treatment Device Technical Report. The aspects which should be covered as part of that TWTD Technical Report may include, but are not limited to, the items listed in table 2 as given in these Guidelines.

2.2 The TWTD Technical Report should be assessed for completeness in respect of the Functional Objectives by the reviewing Administration and further developed by the applicant as considered necessary by the Administration together with such physical surveys as required of the device in production and, as a unit, as installed and in operation. This TWTD Technical

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as considered necessary by the Administration together with such physical surveys as required of the device in production and, as a unit, as installed and in operation. This TWTD Technical Report should thereafter form the basis of the overall approval package of that thermal waste treatment device as an equivalent means to incinerat ion for permitted onboard garbage and waste disposal. Thereafter individual unit certification should be in accordance with the procedures as agreed in respect of the related Functional Objective Unit certification resulting in the issue to each unit of a TWTD Certificate, the form of which is given in the annex, and the approval of that unit's TWTD File.

2.3 These Guidelines cover only the MARPOL Annex VI, prevention of air pollution, aspects related to the use of TWTD. The manufacturer, installer, shipowner and others, as applicable, are responsible for ensuring that all other relevant statutory requirements, together

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aspects related to the use of TWTD. The manufacturer, installer, shipowner and others, as applicable, are responsible for ensuring that all other relevant statutory requirements, together with relevant classification requirements, are complied with as and where appropriate.

2.4 These Guidelines may involve hazardous materials, operations and equipment. These Guidelines do not purport to address the safety aspects associated with the use of thermal waste treatment devices. It is the responsibility of the user of these Guidelines to establish appropriate safety and health practices and determine the applicability of regulatory and classification limitations prior to use, including possible port State limitations.

3 Definitions

Table 1: Definitions

Applicant This may be the device manufacturer or another party in all cases
the applicant is responsible for providing the required information,
performance testing (where required) and subsequent required
ongoing support of the certification
Carbon monoxide (CO) Controlled as an indicator of incomplete oxidation of waste material
otherwise as per the NOx Technical Code 2008

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performance testing (where required) and subsequent required
ongoing support of the certification
Carbon monoxide (CO) Controlled as an indicator of incomplete oxidation of waste material
otherwise as per the NOx Technical Code 2008
Event Record Points Events to be recorded for the purpose of reflecting compliant
operation of the device as installed
Functional Objectives These are the objectives which should be met in order that the
thermal waste treatment device is designed, manufactured,
installed, operated, maintained and serviced such that the required
emission performance is achieved and that as an equivalent means
other uncontrolled pollution streams are not generated
PAH Polycyclic Aromatic Hydrocarbons expressed in terms of
phenanthrene equivalent as defined in the 2021 Guidelines for